Week 11 Notes

STAT 251 SECTION 01

Three distributions behind every test

Every test so far has quietly used **three different distributions**. Most mistakes with tests come from mixing them up, so it is worth laying them side by side before going further.

Take two variables measured on each individual in a sample:

- X, a **quantitative** variable. For example, X = the most weight a college football player can bench press, in pounds.
- Y, a **categorical** variable with two outcomes, recorded as $Y = 1$ for a "success" and $Y = 0$ otherwise. For example, $Y = 1$ if a household has three cell phones.

1. The distribution of the variable. This is how the values of X (or Y) are spread across the whole **population**, one individual at a time.

- X has a population mean μ and a population standard deviation σ. Its shape can be anything: normal, skewed, lumpy. When it is normal, we write

  $$X \sim N(\mu, \sigma) \qquad (\text{normal})$$

  read "X follows a normal distribution with mean μ and standard deviation σ".
- Y has only two values, so its whole distribution is two probabilities:

  $$P(Y = 1) = p, \qquad P(Y = 0) = 1 - p$$

  Recall, this is called a **Bernoulli** distribution, written $Y \sim \text{Bernoulli}(p)$.

2. The sampling distribution of the estimate. We estimate μ with $\bar{x}$ and p with $\hat{p}$, both computed from a sample of size n. A different sample gives a different $\bar{x}$ or $\hat{p}$, and the sampling distribution (Week 8) describes how they vary from sample to sample.

- The sample mean is

  $$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \qquad (\text{normal})$$

  exactly if X is normal, and approximately when $n \geq 30$ whatever the shape of X (the Central Limit Theorem).

- The sample proportion is approximately

$$\hat{p} \sim N\left(p, \sqrt{\frac{p(1-p)}{n}}\right) \qquad \text{(normal)}$$

  when $np \geq 15$ and $n(1-p) \geq 15$.

3. The distribution of the test statistic, if H_0 is true. A test standardizes the estimate, using the **null value** in place of the unknown parameter. If H_0 is true, the result has a distribution we know exactly, so we can say how surprising our sample is:

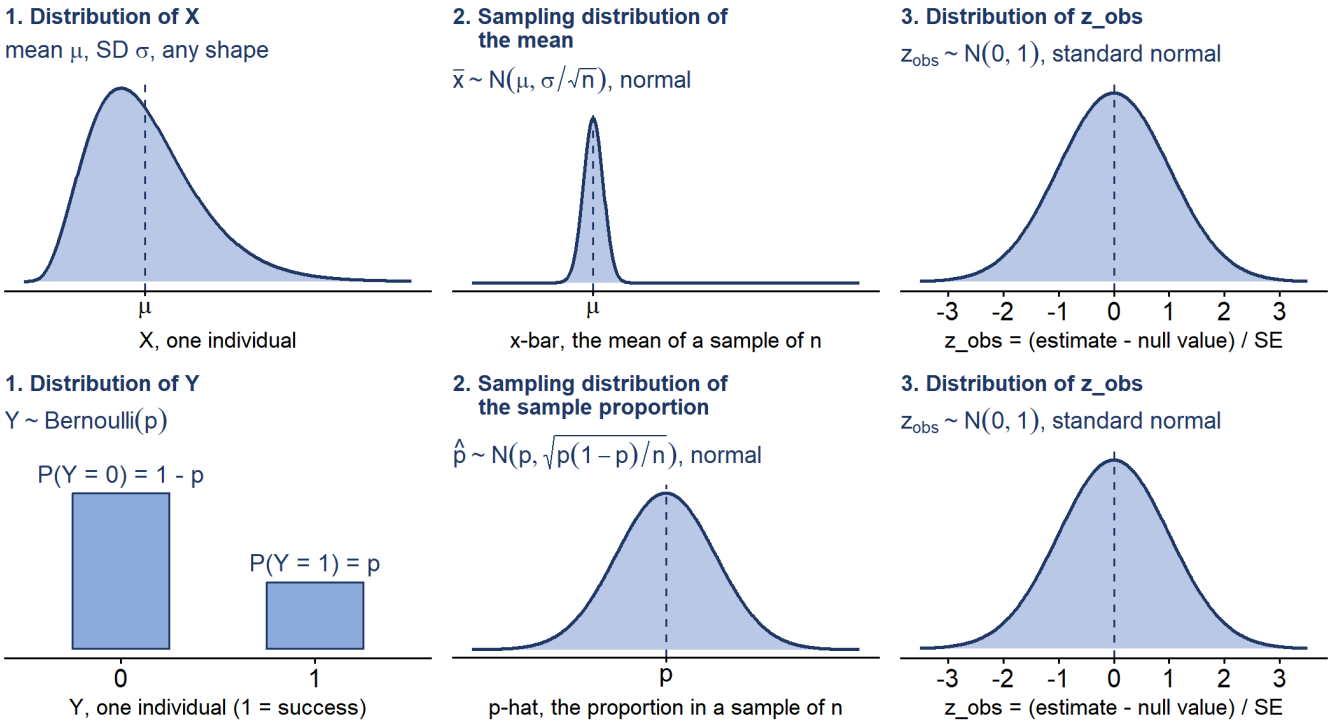
$$z_{obs} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1) \qquad z_{obs} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \sim N(0,1) \qquad \text{(standard normal)}$$

$$t_{obs} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t(n-1) \qquad \text{(Student's } t \text{ with } df = n-1)$$

The subscript "obs" (observed) marks the test statistic computed from our sample. It keeps it apart from a **critical value** such as $Z_{1-\alpha/2}$ (two-sided) or $Z_{1-\alpha}$ (upper-tailed), which comes from the table and not from the data. For example, $Z_{1-\alpha} = Z_{0.95} = 1.645$ when $\alpha = 0.05$.

	Quantitative X: the mean	Categorical Y: the proportion
1. the variable (one individual)	mean μ, SD σ, any shape; if normal, $X \sim N(\mu, \sigma)$ (normal)	$Y \sim \text{Bernoulli}(p)$: $P(Y=1) = p$, $P(Y=0) = 1-p$
2. the estimate (one sample of size n)	$\bar{x} \sim N(\mu, \sigma/\sqrt{n})$ (normal), if X is normal or $n \geq 30$	$\hat{p} \sim N\left(p, \sqrt{p(1-p)/n}\right)$ (normal), if $np \geq 15$ and $n(1-p) \geq 15$
3. the test statistic, if H_0 is true	σ known: $z_{obs} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ $\sim N(0,1)$ (standard normal) σ unknown: $t_{obs} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ $\sim t(n-1)$ (Student's t)	$z_{obs} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \sim N(0,1)$ (standard normal)

Here are the three, side by side, for a quantitative X (top row) and a two-outcome Y (bottom row). The estimate and test-statistic panels are drawn as if H_0 is true.



Three things to notice:

- **The estimate varies much less than the variable.** Panels 1 and 2 of the top row share an axis: one individual's X can land far from μ, but the mean of n of them stays close. That is the $\sqrt{n}$ in $\sigma/\sqrt{n}$.
- **The variable does not have to be normal.** X is skewed, and Y is two bars, nothing like a bell. Their estimates $\bar{x}$ and $\hat{p}$ are still close to normal when n is large (the Central Limit Theorem).
- **The last panel is the same in both rows.** Standardizing removes the units and the null value, so every z_{obs} ends on the same standard normal curve, and one z-table serves them all. A t_{obs} ends on a t curve instead, which depends only on df.

The **P-value is always an area under the third distribution**: the distribution of the test statistic when H_0 is true.

Why the test statistic is standard normal when H_0 is true

Recall the **z-score** from Week 6. To standardize a value, subtract the **mean of its distribution** and divide by the **standard deviation of its distribution**:

$$z = \frac{\text{value} - \text{mean}}{\text{SD}}$$

If the value comes from a normal distribution, its z-score follows $N(0, 1)$ (standard normal). Every normal curve turns into the same standard normal curve once we measure in SDs from its centre.

Now look at the test statistic for a mean, one piece at a time.

- The value we standardize is $\bar{x}$. Its distribution is $\bar{x} \sim N(\mu, \sigma/\sqrt{n})$.
- **If H_0 is true**, then $\mu = \mu_0$, so $\bar{x} \sim N(\mu_0, \sigma/\sqrt{n})$. The mean of its distribution is μ_0, and its SD is $\sigma/\sqrt{n}$.
- So the test statistic is exactly the z-score of $\bar{x}$:

$$z_{obs} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{\text{value} - \text{its mean if } H_0 \text{ is true}}{\text{its SD}} \sim N(0,1) \qquad (\text{standard normal})$$

The proportion works the same way. If H_0 is true, $p = p_0$, so $\hat{p} \sim N\left(p_0, \sqrt{p_0(1-p_0)/n}\right)$, and

$$z_{obs} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \sim N(0,1) \qquad (\text{standard normal})$$

is the z-score of $\hat{p}$. This is why the standard error in a test uses p_0 and not $\hat{p}$: if H_0 is true, $\sqrt{p_0(1-p_0)/n}$ **is** the SD of $\hat{p}$.

A number to hold on to: the bench press. *From OpenStax Introductory Statistics 2e, Section 9.5, Example 9.15 (https://openstax.org/books/introductory-statistics-2e/pages/9-5-additional-information-and-full-hypothesis-test-examples).*

> A college football coach records the mean weight that his players can bench press as **275 pounds**, with a standard deviation of $\sigma = 55$ pounds. Three of the players think the mean is more than that. They ask **30** of their teammates for their estimated maximum bench press. The answers range from 205 to 385 pounds; the weights, with the number of players giving each in parentheses, are 205 (3), 215 (3), 225 (1), 241 (2), 252 (2), 265 (2), 275 (2), 313 (2), 316 (5), 338 (2), 341 (1), 345 (2), 368 (2), 385 (1). Using a 2.5% significance level, test whether the mean bench press is more than 275 pounds.

The hypotheses are $H_0 : \mu = 275$ and $H_A : \mu > 275$, and the 30 weights add up to 8,585 pounds, so $\bar{x} = 8{,}585/30 \approx 286.17$ pounds.

- **If H_0 is true**, the mean of 30 players follows

$$\bar{x} \sim N\left(275, \frac{55}{\sqrt{30}}\right) = N(275, 10.04) \qquad (\text{normal})$$

- The players' mean, $\bar{x} = 286.17$, sits this many SDs above the centre of that curve:

$$\frac{286.17 - 275}{10.04} \approx 1.11$$

- That z-score **is** the test statistic: $z_{obs} = 1.11$, and if H_0 is true, $z_{obs} \sim N(0, 1)$ (standard normal).

To finish the test, the P-value is $P(Z \geq 1.11 \mid H_0 \text{ true}) = 1 - 0.8665 = 0.1335$ (the book's calculator, without rounding z_{obs}, gives 0.1331). Since $0.1335 \geq 0.025$, **do not reject** H_0: at the 2.5% level, the data are not convincing evidence that the mean bench press is more than 275 pounds.

What if H_0 is false? Then the true mean μ is not μ_0, and subtracting μ_0 subtracts the wrong centre. z_{obs} is still normal with SD 1, but its centre moves to $(\mu - \mu_0)/(\sigma/\sqrt{n})$, away from 0. That is why a z_{obs} far from 0 is evidence against H_0: it is the kind of value we expect when the curve has shifted. (This shifted curve comes back in the section on power.)

Why t_{obs} is not a z-score. With σ unknown, we divide by $s/\sqrt{n}$ instead. But s is an estimate that changes from sample to sample, so we are not dividing by the true SD of $\bar{x}$. That extra uncertainty spreads the statistic out, and if H_0 is true it follows

$$t_{obs} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t(n-1) \qquad (\text{Student's } t \text{ with } df = n - 1)$$

which has heavier tails than $N(0, 1)$ and gets closer to it as n grows.

Composite hypotheses

So far every null hypothesis has named a single value of the parameter: $H_0 : \mu = 275$, $H_0 : p = 0.5$. A hypothesis that **completely specifies** the parameter like this is called a **simple** hypothesis.

Often, though, the "nothing interesting" story does not pin the parameter down to one value. If the claim is that a glass conducts electricity **better** than 1 (the glass example below), then a mean of exactly 1 is not the only way for the claim to be wrong: 0.99, 0.95 or 0.80 would be just as disappointing. The null has to include all of them, $H_0 : \mu \leq 1$. A hypothesis that allows a whole **range** of values, because the value of the parameter is not completely specified, is called a **composite** hypothesis.

If H_A is	the simple null is	the composite null is
$\theta > \theta_0$ (upper-tailed)	$H_0 : \theta = \theta_0$	$H_0 : \theta \leq \theta_0$
$\theta < \theta_0$ (lower-tailed)	$H_0 : \theta = \theta_0$	$H_0 : \theta \geq \theta_0$
$\theta \neq \theta_0$ (two-sided)	$H_0 : \theta = \theta_0$	(none)

Here θ stands for whichever parameter we are testing, μ or p. Two rules follow from this (summarized in OpenStax Table 9.1):

- H_0 **always contains the equals sign** ($=$, $\leq$ or $\geq$). H_A **never does** ($\neq$, $>$ or $<$).
- Together, H_0 and H_A cover **every** possible value of the parameter. With $H_A : \mu > 1$, the composite null $H_0 : \mu \leq 1$ is "everything else".

When is a composite null the right one?

A composite null is the honest way to write a **one-sided** claim, where the "nothing interesting" story is a whole range of values, not one point. Listen for the words:

- "**at least**", "**no less than**" $\rightarrow H_0$ uses $\geq$;
- "**at most**", "**no more than**" $\rightarrow H_0$ uses $\leq$.

Some examples (from OpenStax Section 9.1 (https://openstax.org/books/introductory-statistics-2e/pages/9-1-null-and-alternative-hypotheses) and Section 9.2 (https://openstax.org/books/introductory-statistics-2e/pages/9-2-outcomes-and-the-type-i-and-type-ii-errors)):

For each one, name the parameter in words, then write H_0 and H_A. (The book gives each claim in one sentence; the setup around it is ours, so there is a story to test.)

Voter turnout (Example 9.1). A county election official reports that **no more than 30%** of the registered voters in Santa Clara County voted in the primary election. A local newspaper suspects turnout was **higher**, and surveys a random sample of registered voters.

- *Parameter:* p = the proportion of **all** registered voters in the county who voted in the primary.
- *Hypotheses:* the official's figure, "30% or less", is the null: $H_0 : p \leq 0.30$. The newspaper's suspicion is the alternative: $H_A : p > 0.30$. Composite, **upper-tailed**, tested at $p_0 = 0.30$.

Time to graduate (Example 9.3). College students are often told a degree takes **five years**, on average. An advising office believes its students finish **faster**, and records how long a random sample of recent graduates took to graduate.

- *Parameter:* μ = the mean time to graduate for **all** of the college's students.
- *Hypotheses:* the office's claim, "less than 5 years", is the alternative, $H_A : \mu < 5$. Taking exactly 5 years or longer are both "not less than 5", so $H_0 : \mu \geq 5$. Composite, **lower-tailed**, tested at $\mu_0 = 5$.

A cure-rate claim (Example 9.8). A drug company advertises that its experimental drug cures **at least 75%** of men with prostate cancer. An independent research group suspects the true cure rate is **lower**, and follows a random sample of patients who took the drug.

- *Parameter:* p = the proportion of **all** men with prostate cancer who are cured by the drug.
- *Hypotheses:* the company's claim, "75% or more", is the null: $H_0 : p \geq 0.75$. The research group's suspicion is the alternative: $H_A : p < 0.75$. Composite, **lower-tailed**, tested at $p_0 = 0.75$.

A two-sided test **never** has a composite null: if H_A is $\theta \neq \theta_0$, the only value left for H_0 is $\theta = \theta_0$.

Example: the conductivity of glass (a composite null)

From OpenStax Introductory Statistics 2e, Section 9.5, Example 9.20 (https://openstax.org/books/introductory-statistics-2e/pages/9-5-additional-information-and-full-hypothesis-test-examples).

> The National Institute of Standards and Technology provides exact data on conductivity properties of materials. Following are conductivity measurements for **11** randomly selected pieces of a particular type of glass:
>
> 1.11, 1.07, 1.11, 1.07, 1.12, 1.08, 0.98, 0.98, 1.02, 0.95, 0.95
>
> Is there convincing evidence that the average conductivity of this type of glass is **greater than one**? Use a significance level of 0.05. Assume the population is normal.

Only the data are given, so σ is unknown and this is a t-test.

Question: At $\alpha = 0.05$, is the mean conductivity of this glass greater than 1?

1. **Assumptions.** A random sample of 11 pieces. n is small, but the population is normal, so the test works. σ is unknown, so we use a t-test.
2. **Hypotheses.** μ = the mean conductivity of all glass of this type. The claim "greater than 1" is the alternative, $H_A : \mu > 1$, and the null is everything else, a composite null: $H_0 : \mu$

≤ 1. The test is **upper-tailed**, and it is done at the boundary, $\mu_0 = 1$ (the next section explains why).

3. **Test statistic.** From the data, $\bar{x} = 1.04$ and $s = 0.0659$.

$$SE = \frac{s}{\sqrt{n}} = \frac{0.0659}{\sqrt{11}} \approx 0.0199$$

$$t_{obs} = \frac{\bar{x} - \mu_0}{SE} = \frac{1.04 - 1}{0.0199} \approx 2.01, \qquad df = 11 - 1 = 10$$

4. **P-value.** In row 10 of the t-table (misc/t_table.html), $1.812 < 2.01 < 2.228$. Those columns have upper-tail areas 0.05 and 0.025, so

$$0.025 < P(T \geq 2.01 \mid \mu = 1) < 0.05$$

   Software gives 0.036.

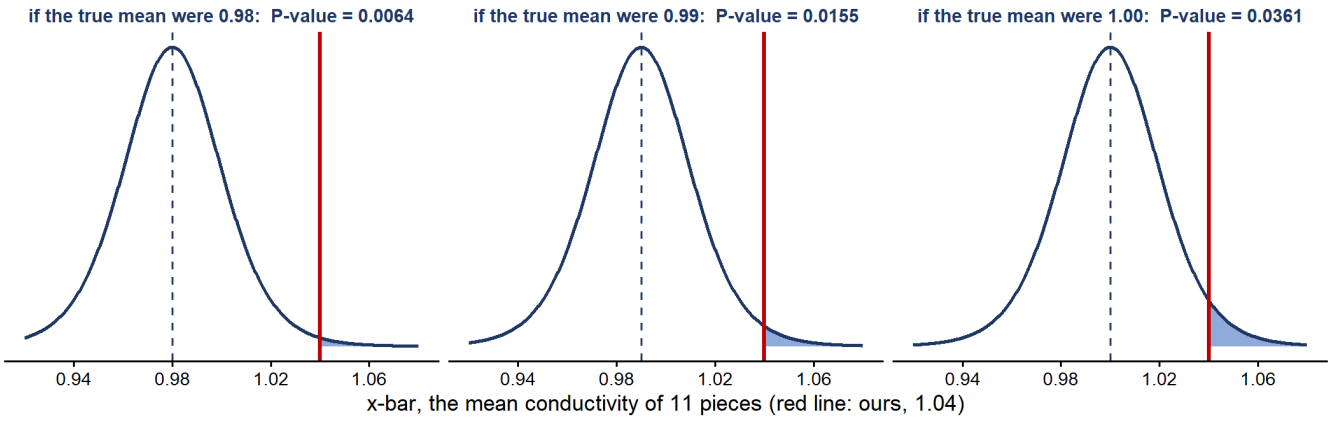
5. **Decision.** The P-value is below $\alpha = 0.05$: **reject** H_0. The data are convincing evidence that the mean conductivity of this glass is greater than 1.

How do we test a composite null?

There is a catch. To compute a test statistic we need **one** null value, but $H_0 : \mu \leq 1$ has infinitely many: 1, 0.99, 0.98, and so on. Which one goes into the formula?

Answer: the boundary value, $\mu_0 = 1$. The test statistic and P-value are computed exactly as if the null were $H_0 : \mu = 1$, as in the glass example. Here is why. The glass data have $\bar{x} = 1.04$. Look at the P-value for three values that are all inside $H_0 : \mu \leq 1$:

if the true mean were	t_{obs}	P-value $P(T \geq t_{obs})$
$\mu = 0.98$	$(1.04 - 0.98)/0.0199 = 3.02$	0.0064
$\mu = 0.99$	$(1.04 - 0.99)/0.0199 = 2.51$	0.0155
$\mu = 1.00$ (the boundary)	$(1.04 - 1.00)/0.0199 = 2.01$	0.0361



The farther the true mean is **below** 1, the less likely a sample mean as high as 1.04 becomes, so the P-value shrinks. The **largest** P-value in H_0 is at the boundary, $\mu = 1$. So:

- If we reject at the boundary (P-value $< \alpha$ there), the P-value is even smaller for every other value in H_0, so we reject all of H_0 at once.
- The boundary is the **hardest** case to reject, so testing there is the cautious choice. It also means the chance of wrongly rejecting is **at most** α, whichever value in H_0 is the truth.

That is also why many researchers write $H_0 : \mu = 1$ even when they mean $H_0 : \mu \leq 1$: **the calculation and the decision are identical.** The composite version says more honestly what we would conclude if we do not reject: "the mean could be 1 **or less**".

Example: the baker's bread (a composite null)

From OpenStax Introductory Statistics 2e, Section 9.4, Example 9.9 (https://openstax.org/books/introductory-statistics-2e/pages/9-4-rare-events-the-sample-decision-and-conclusion).

> Suppose a baker claims that his bread height is **more than 15 cm**, on average. Several of his customers do not believe him. To persuade his customers that he is right, the baker decides to do a hypothesis test. He bakes **10** loaves of bread. The mean height of the sample loaves is **17 cm**. The baker knows from baking hundreds of loaves of bread that the standard deviation for the height is $\sigma = 0.5$ cm, and the distribution of heights is normal.

The book gives no significance level, so we use the usual $\alpha = 0.05$.

Question: At $\alpha = 0.05$, do the data back up the baker's claim?

1. **Assumptions.** We treat the 10 loaves as a random sample of his bread. The heights are normal, so $n = 10$ is enough. $\sigma = 0.5$ is known, so we use a z-test.
2. **Hypotheses.** μ = the mean height of all of the baker's loaves. The claim "more than 15 cm" is the alternative, $H_A : \mu > 15$. The customers' view is everything else, "15 cm **or less**", so the null is composite: $H_0 : \mu \leq 15$. The test is **upper-tailed**, done at the boundary $\mu_0 = 15$.
3. **Test statistic.**

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{0.5}{\sqrt{10}} \approx 0.1581$$

$$z_{obs} = \frac{\bar{x} - \mu_0}{SE} = \frac{17 - 15}{0.1581} \approx 12.65$$

   The sample mean is more than 12 standard errors above 15 cm.
4. **P-value.** $P(Z \geq 12.65 \mid \mu = 15)$ is far off the end of the z-table: **essentially 0**. And that is the largest P-value for any value in H_0; if the true mean were below 15, a sample mean of 17 would be even less likely.
5. **Decision.** P-value $\approx 0 < 0.05$: **reject** H_0. The data are convincing evidence that the baker's loaves average more than 15 cm.

Try it out

From OpenStax Introductory Statistics 2e, Section 9.4, Try It 9.9 (https://openstax.org/books/introductory-statistics-2e/pages/9-4-rare-events-the-sample-decision-and-conclusion).

> A normal distribution has a standard deviation of **1**. We want to verify a claim that the mean is **greater than 12**. A sample of **36** is taken with a sample mean of **12.5**. The hypotheses are $H_0 : \mu \leq 12$ and $H_A : \mu > 12$, and the P-value is 0.0013. Draw a graph that shows the P-value.

Check the book's P-value by running the test yourself, and decide at $\alpha = 0.05$ (the book gives no α).

Answer: $H_0 : \mu \leq 12$ and $H_A : \mu > 12$, tested at the boundary $\mu_0 = 12$. $SE = 1/\sqrt{36} = 1/6 \approx 0.1667$, so $z_{obs} = (12.5 - 12)/(1/6) = 3.00$. The P-value is $P(Z \geq 3.00 \mid \mu = 12) = 1 - 0.9987 = 0.0013 < 0.05$: reject H_0. The mean is greater than 12.

Confidence intervals and two-sided tests

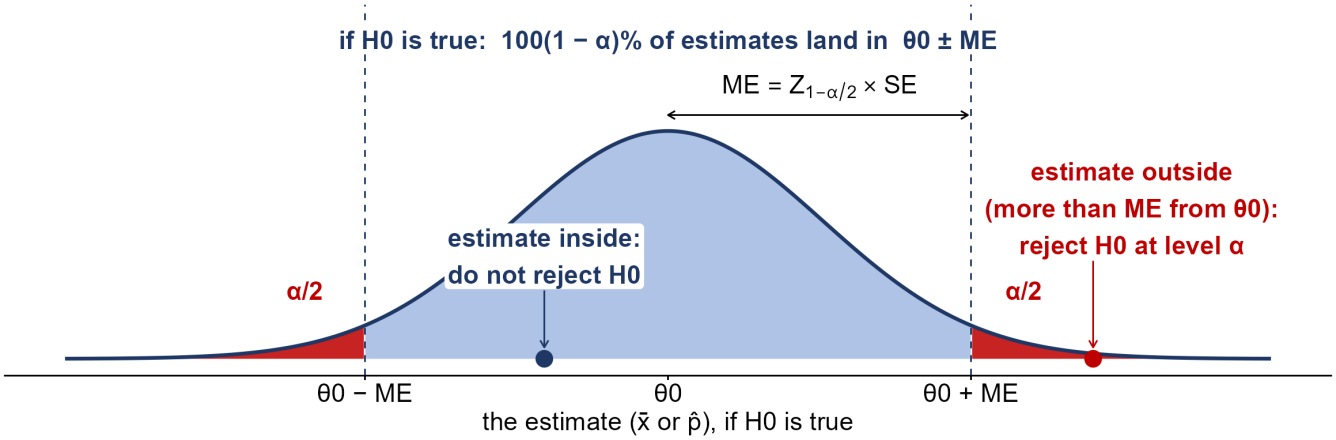
This is the decision rule from Monday seen as an interval. Write θ for the parameter (μ or p) and $\hat{\theta}$ for its estimate ($\bar{x}$ or $\hat{p}$).

The null sets an interval. If $H_0 : \theta = \theta_0$ is true, the sampling distribution of the estimate is centred at θ_0, and $100(1-\alpha)\%$ of all estimates land within one **margin of error** of it:

$$\theta_0 \pm ME, \qquad ME = Z_{1-\alpha/2} \times SE$$

(with $t_{n-1,\,1-\alpha/2}$ in place of $Z_{1-\alpha/2}$ when σ is unknown). The two tails outside this interval hold the remaining α, $\alpha/2$ on each side.

An estimate outside it is evidence against H_0. If our estimate lands more than ME from θ_0, it is in a region that holds only $100\alpha\%$ of the estimates H_0 predicts. That is the same as $|z_{obs}| > Z_{1-\alpha/2}$, so a two-sided test **rejects H_0 at level α**. An estimate inside the interval gives no such evidence: **do not reject**.



The confidence interval says the same thing. "$\hat{\theta}$ is more than ME from θ_0" is the same statement as "θ_0 is more than ME from $\hat{\theta}$", that is, θ_0 lies outside the $100(1-\alpha)\%$ confidence interval $\hat{\theta} \pm ME$. So a 95% confidence interval is a two-sided test at $\alpha = 0.05$ of every value at once: values outside it would be rejected, values inside it would not.

Two cautions. The match is for **two-sided** tests; a one-sided test puts all of α in one tail. And for a **proportion** the match is close but not exact, because the test's SE uses p_0 while the interval's uses $\hat{p}$.

Types of errors

A test ends in a decision, reject H_0 or do not reject it, and it makes that decision **without ever learning the truth**. So there are four ways it can turn out (OpenStax Table 9.2):

	H_0 is actually true	H_0 is actually false
Do not reject H_0	correct	**Type II error** probability β
Reject H_0	**Type I error** probability α	correct probability $1-\beta$, the **power**

- A **Type I error** is rejecting H_0 when it is true: a **false alarm**. Its probability is the significance level,

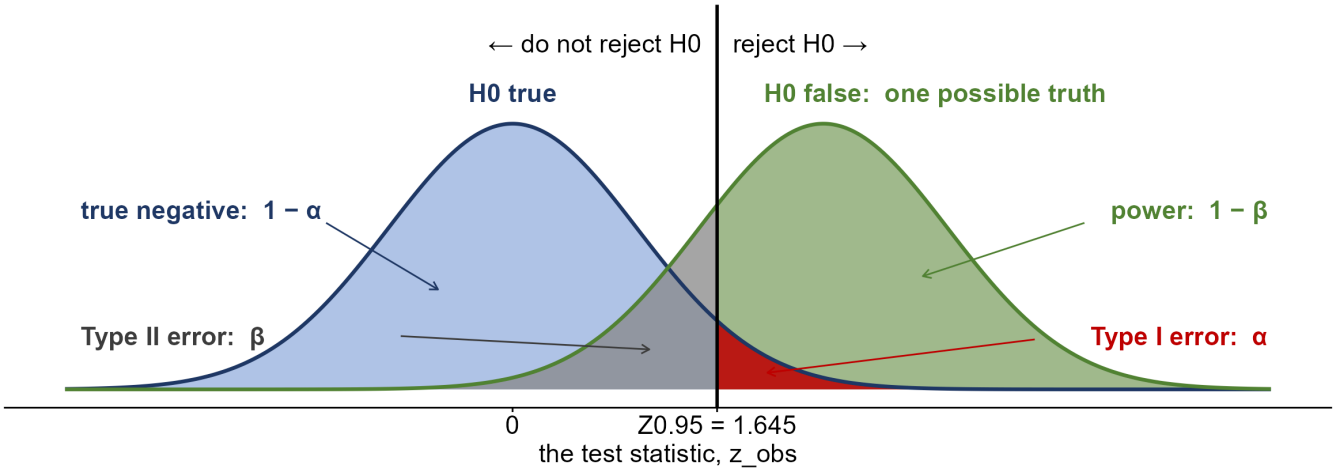
$$\alpha = P(\text{Type I error}) = P(\text{reject } H_0 \mid H_0 \text{ true})$$

  We choose α before collecting the data, so we decide in advance how often we are willing to raise a false alarm.
- A **Type II error** is not rejecting H_0 when it is false: a **missed effect**. Its probability is

$$\beta = P(\text{Type II error}) = P(\text{do not reject } H_0 \mid H_0 \text{ false})$$

The same four outcomes can be read off the distributions of the test statistic. The picture below is a generic upper-tailed z-test at $\alpha = 0.05$: one curve shows where z_{obs} lands if H_0 is true, and the critical value $Z_{0.95} = 1.645$ splits "do not reject" (left) from "reject" (right).

The other curve needs care. "H_0 is false" is not one model: the parameter could take any of many values outside H_0, and each one gives its own curve. The picture draws just **one possible truth**, as the power example below does with a true mean of 16.0 seconds. A test never tells us which of those values is the truth; rejecting H_0 only tells us that the null model is unlikely.



- **Under the "H_0 true" curve** (the world where H_0 is true): the area left of the line is a correct "do not reject", a **true negative**, with probability $1 - \alpha$; the tail to the right is a **Type I error**, with probability α.
- **Under the "one possible truth" curve** (a world where H_0 is false): the area left of the line is a **Type II error**, with probability β; the area to the right is a correct rejection, the **power**, $1 - \beta$.
- Sliding the line to the right shrinks α but grows β, and sliding it left does the opposite. That is the trade-off between the two errors for a fixed sample size.

Back to the courtroom from the Week 10 notes, where H_0 is "the defendant is innocent". A Type I error convicts an innocent person. A Type II error lets a guilty person go free. The legal system sets a high bar, "beyond reasonable doubt", because it considers the first error worse; the price is more of the second.

Which error is worse depends on the situation. In each example below, write H_0 down first; the errors follow from it.

Example: errors in context

From OpenStax Introductory Statistics 2e, Section 9.2, Examples 9.5, 9.7 and 9.8 (https://openstax.org/books/introductory-statistics-2e/pages/9-2-outcomes-and-the-type-i-and-type-ii-errors).

Navah's rock climbing equipment (Example 9.5)

Navah is a rock climber. Before a climb she has to decide whether her ropes, harness and anchors are safe to trust. Because she has used the gear before, her starting assumption, the null hypothesis, is that it is fine. (The book states only the null hypothesis; this setup is ours.)

> Suppose the null hypothesis, H_0, is: **Navah's rock climbing equipment is safe.** State the Type I and Type II errors. Which error has the greater consequence?

- *Type I error* (reject a true H_0): Navah thinks her rock climbing equipment may **not** be safe when, in fact, it really is safe. She retires good gear.
- *Type II error* (do not reject a false H_0): Navah thinks her rock climbing equipment **may be safe** when, in fact, it is not safe. She goes ahead and climbs on it.
- α = the probability that Navah thinks her equipment may not be safe when it really is; β = the probability that she thinks it may be safe when it is not.
- The **Type II error** has the greater consequence: if Navah thinks her equipment is safe, she will use it, and it can fail. Here we want β small, even at the cost of more false alarms.

Genetic Labs (Example 9.7)

> A company called **Genetic Labs** claims to be able to increase the likelihood that a pregnancy will result in a **male** being born. Statisticians want to test the claim. Suppose that the null hypothesis, H_0, is: **Genetic Labs has no effect on sex outcome.** Describe the Type I and Type II errors, and say which has the greater consequence.

- *Type I error:* we conclude that Genetic Labs **does** influence the sex outcome, when in fact it has no effect. Its probability is α.
- *Type II error:* we conclude that Genetic Labs does **not** influence the sex outcome of a pregnancy when, in fact, it does. Its probability is β.
- The **Type I error** has the greater consequence: people would pay for Genetic Labs' product in the hope of having a boy, when it does nothing. Here we want α small.

A cancer drug (Example 9.8, a composite null)

> A certain experimental drug claims a cure rate of **at least 75%** for males with prostate cancer. Describe both the Type I and Type II errors in context. Which error is the more serious?

The claim "at least 75%" is the null, a composite one: $H_0 : p \geq 0.75$, and a skeptic tests it against $H_A : p < 0.75$.

- *Type I error:* a cancer patient believes the cure rate for the drug is **less than 75%** when it actually is at least 75%. He may turn down a drug that works.
- *Type II error:* a cancer patient believes the experimental drug has **at least a 75%** cure rate when it has a cure rate that is less than 75%. He relies on a drug that works less well than claimed.
- The **Type II error** is more serious: a patient and doctor who believe the drug works at least 75% of the time will likely choose it as the treatment.

Practice: cell phone owners

From OpenStax Introductory Statistics 2e, Section 9.5, Try It 9.18 (https://openstax.org/books/introductory-statistics-2e/pages/9-5-additional-information-and-full-hypothesis-test-examples).

> Marketers believe that **92%** of adults in the United States own a cell phone. A cell phone manufacturer believes that number is actually **lower**. **200** American adults are surveyed, of which **174** report having cell phones. Use a 5% level of significance. State the null and alternative hypotheses, find the P-value, state your conclusion, and identify the Type I and Type II errors.

The book leaves this one as an exercise; the solution below is ours.

1. **Assumptions.** A random sample of adults: $np_0 = 200(0.92) = 184 \geq 15$ and $n(1 - p_0) = 200(0.08) = 16 \geq 15$.
2. **Hypotheses.** p = the proportion of all U.S. adults who own a cell phone. The marketers' figure is "92% or more", so $H_0 : p \geq 0.92$ (composite), and the manufacturer's suspicion is $H_A : p < 0.92$: **lower-tailed**, tested at the boundary $p_0 = 0.92$.
3. **Test statistic.** $\hat{p} = 174/200 = 0.87$.

$$SE = \sqrt{\frac{0.92 \times 0.08}{200}} \approx 0.0192, \qquad z_{obs} = \frac{0.87 - 0.92}{0.0192} \approx -2.60$$

4. **P-value.** $P(Z \leq -2.60 \mid p = 0.92) = 0.0047$.
5. **Decision.** $0.0047 < 0.05$: **reject** H_0. The data are convincing evidence that fewer than 92% of U.S. adults own a cell phone.

The errors. A **Type I error** would be concluding that fewer than 92% own a cell phone when really at least 92% do. A **Type II error** would be concluding that the 92% figure could be right when really fewer own one. Since we rejected H_0, the only error we could have made here is a Type I error.

Statistical power

The **power** of a test is its chance of catching an effect that is really there:

$$\text{Power} = P(\text{reject } H_0 \mid H_0 \text{ false}) = 1 - \beta$$

A test with low power can easily miss a real effect and end with "do not reject", which is one more reason that "do not reject" never means "H_0 is true". We want power close to 1.

Power has a catch: "H_0 is false" is not one situation. It depends on **how** false. A test finds a big effect more easily than a small one, so a power calculation always starts by assuming a particular true value of the parameter.

Example: how often would we catch the goggles working?

Return to Jeffrey and his goggles from Lecture 24, *from OpenStax Introductory Statistics 2e, Section 9.5, Example 9.14 (https://openstax.org/books/introductory-statistics-2e/pages/9-5-additional-information-and-full-hypothesis-test-examples)*:

> Jeffrey, as an eight-year old, established a mean time of **16.43 seconds** for swimming the 25-yard freestyle, with a standard deviation of $\sigma = 0.8$ seconds. His dad, Frank, thought that Jeffrey could swim the 25-yard freestyle faster using goggles. Frank bought Jeffrey a new pair of expensive goggles and timed Jeffrey for **15** 25-yard freestyle swims. For the 15 swims, Jeffrey's mean time was 16 seconds. Frank thought that the goggles helped Jeffrey to swim faster than the 16.43 seconds. Conduct a hypothesis test using a preset $\alpha = 0.05$. Assume that the swim times for the 25-yard freestyle are normal.

Frank's test is $H_0 : \mu = 16.43$ against $H_A : \mu < 16.43$ at $\alpha = 0.05$, a lower-tailed z-test (Week 10 works it: $z_{obs} = -2.08$, reject H_0). Power asks a different question, one Frank could answer **before** timing a single swim.

Suppose the goggles **really do** help, and Jeffrey's true mean in them is **16.0 seconds**. (This value is our assumption; the book does not give one.) Before Frank collects any data, what is the chance his test will reject H_0?

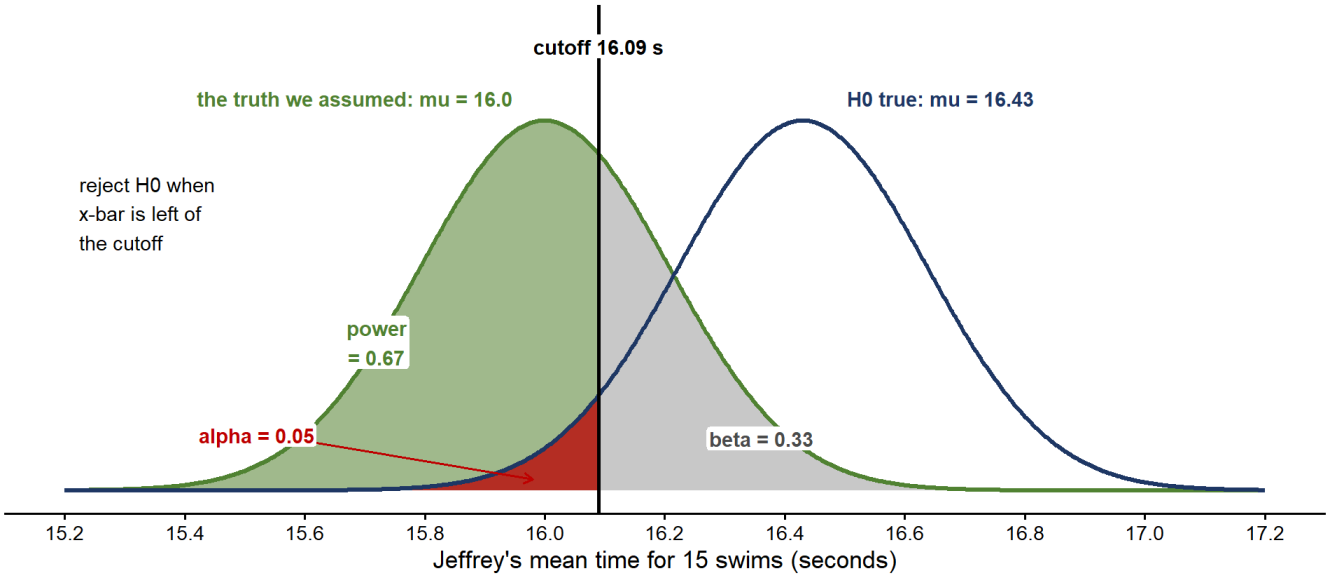
1. **Find the cutoff.** A lower-tailed test rejects when $z_{obs} \le Z_\alpha$, the lower-tail critical value. For $\alpha = 0.05$, $Z_{0.05} = -1.645$. In seconds, with $SE = 0.8/\sqrt{15} \approx 0.2066$, that is a sample mean of

   $$\bar{x} \le 16.43 - 1.645 \times 0.2066 \approx 16.09 \text{ seconds}$$

   The cutoff comes from H_0 alone. If the true mean were 16.43, a sample mean this low would happen only 5% of the time; that is α.
2. **Switch to the truth.** If the true mean is 16.0, the sample mean has a normal distribution centred at **16.0**, with the same SE. The power is the chance that it lands below the cutoff:

   $$\text{Power} = P(\bar{x} \le 16.09 \mid \mu = 16.0) = P\left(Z \le \frac{16.09 - 16.0}{0.2066}\right) = P(Z \le 0.44) = 0.67$$

3. **Interpret.** Even if the goggles really cut Jeffrey's mean to 16.0 seconds, a test with 15 swims catches it only about 67% of the time. The other $\beta \approx 0.33$ of the time, Frank's test misses it: a **Type II error**.

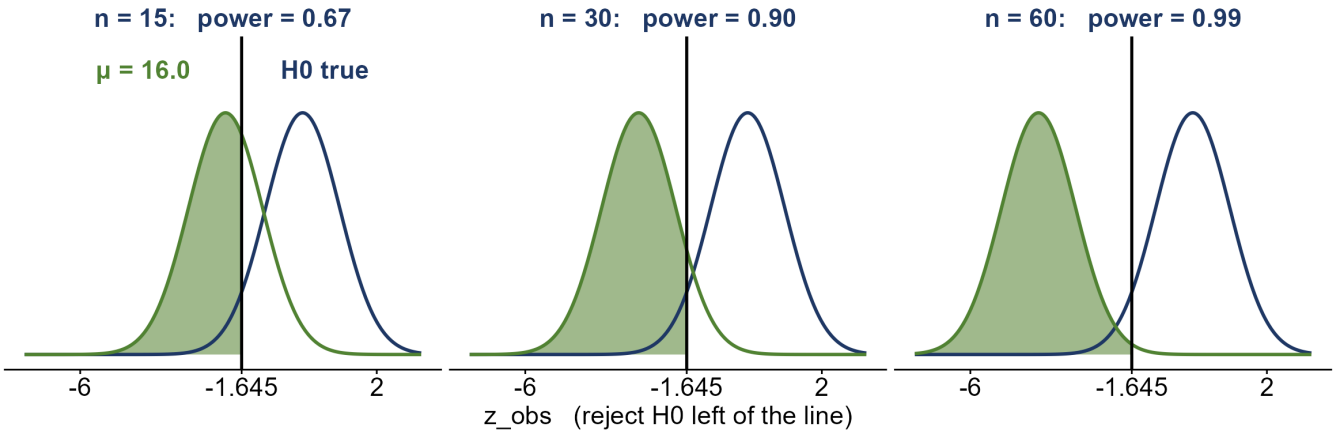


Read the picture from the cutoff line. Everything to its left is "reject H_0". Under the H_0 curve, that area is α, the false alarms. Under the true curve, it is the power. What is left of the true curve, to the right of the line, is β, the misses.

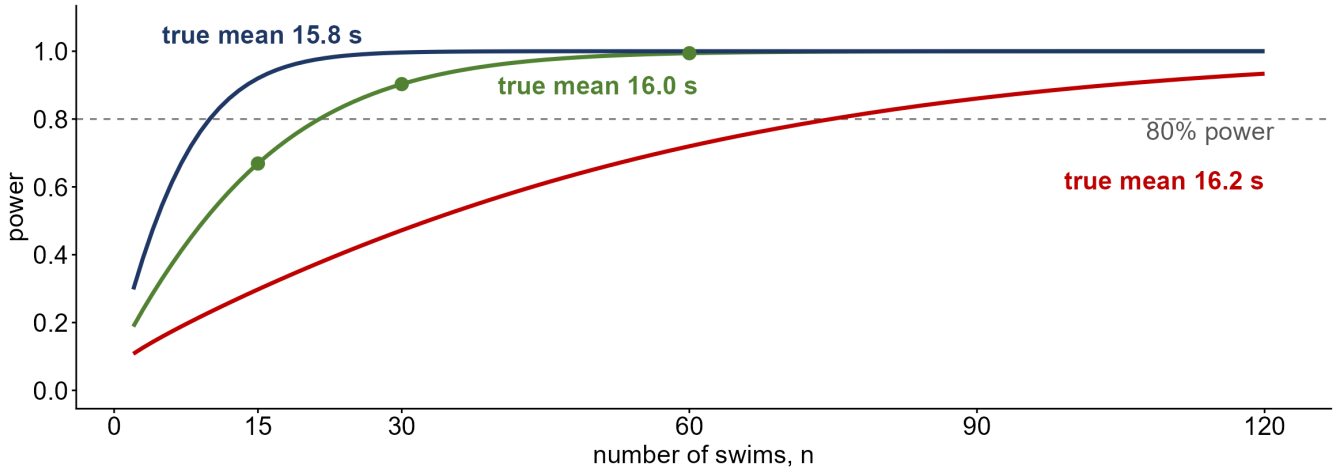
The same test with more swims. Now look at it on the scale of the test statistic, $z_{obs} = (\bar{x} - 16.43)/(0.8/\sqrt{n})$. If H_0 is true, $z_{obs} \sim N(0,1)$ for **every** n, and the cutoff stays at $Z_{0.05} = -1.645$: neither depends on the sample size. But if the true mean is 16.0 seconds, z_{obs} is centred at

$$\frac{16.0 - 16.43}{0.8/\sqrt{n}} = -2.08, \quad -2.94, \quad -4.16 \qquad \text{for } n = 15,\ 30,\ 60$$

The same 0.43-second improvement is divided by a smaller and smaller SE, so the curve for this possible truth slides away from 0 and more of it falls past the cutoff:



Doing this for every sample size gives a **power curve**. Each line below is one possible true mean:



- Power climbs toward 1 as n grows, for any true mean outside H_0.
- The smaller the real effect, the more data it takes. A common target is **80% power**: with a true mean of 16.0 seconds that takes $n = 22$ swims; with 15.8 seconds, $n = 10$; with 16.2 seconds, $n = 75$.
- This is how a study is planned: pick the smallest effect worth detecting, then choose n so the power for that effect is high enough.

What raises the power?

Each change below starts from the goggles test ($n = 15$, $\alpha = 0.05$, true mean 16.0 seconds) and changes **one** thing:

Change	Power
none (the example above)	0.67
more swims: $n = 30$	0.90
more swims: $n = 60$	0.99
smaller $\alpha = 0.01$	0.40
larger $\alpha = 0.10$	0.79
smaller effect: true mean 16.2 s	0.30
bigger effect: true mean 15.8 s	0.92

- **A bigger sample** is the usual fix. More data shrinks $SE = \sigma/\sqrt{n}$, so the two curves overlap less. With 30 swims, $SE \approx 0.1461$, the cutoff moves to $16.43 - 1.645(0.1461) \approx 16.19$, and the power is $P(Z \leq (16.19 - 16)/0.1461) = P(Z \leq 1.30) = 0.90$.
- **A larger α** moves the cutoff toward the alternative and raises the power, but only by allowing more Type I errors. For a fixed sample size, α and β pull against each other: making one smaller makes the other larger.
- **A bigger true effect** is easier to detect. We cannot control it, but it matters when planning: a study meant to find a small effect needs a large sample.
- **Less variability** (a smaller σ) works like a bigger sample, since it also shrinks SE.

In practice, when the power is too low, statisticians usually **increase the sample size** while keeping α the same.

Try it out

- *(OpenStax Chapter 9 Practice (https://openstax.org/books/introductory-statistics-2e/pages/9-practice), Section 9.2, Exercise 17.)* The power of a test is 0.981. What is the probability

of a Type II error? *Answer:* $\beta = 1 - 0.981 = 0.019$.

- *(OpenStax Chapter 9 Practice (https://openstax.org/books/introductory-statistics-2e/pages/9-practice), Section 9.2, Exercise 19.)* A microbiologist is testing a water sample for E. coli. Suppose the null hypothesis, H_0, is: the sample does not contain E. coli. The probability that the sample does not contain E. coli, but the microbiologist thinks it does, is 0.012. The probability that the sample does contain E. coli, but the microbiologist thinks it does not, is 0.002. What is the power of this test? *Answer:* the first probability is $\alpha = 0.012$ (a false alarm) and the second is $\beta = 0.002$ (a miss). Power $= 1 - \beta = 0.998$.
- *(OpenStax Section 9.2, Try It 9.6 (https://openstax.org/books/introductory-statistics-2e/pages/9-2-outcomes-and-the-type-i-and-type-ii-errors).)* Suppose the null hypothesis, H_0, is: a patient is not sick. Which type of error has the greater consequence, Type I or Type II? *Answer:* a Type II error, deciding the patient is not sick when they are. A sick patient goes untreated.

Parametric and non-parametric tests

Every test so far has been **parametric**: it assumes a model for the population, written in terms of its parameters, and takes its P-value from the distribution that model implies. Here are the assumptions of each one:

Test	The population	The sample	Then, if H0 is true
every test so far	(any)	a random sample (or a randomized experiment) of independent observations	
z-test for p	each individual is a success or a failure: $Y \sim \text{Bernoulli}(p)$, so the number of successes in the sample is $\text{Binomial}(n, p)$	$np_0 \geq 15$ and $n(1-p_0) \geq 15$, so that $\hat{p}$ is approximately normal	$z_{obs} \sim N(0,1)$
z-test for μ (σ known)	approximately normal, $X \sim N(\mu, \sigma)$, with σ known	any n if the population is normal; otherwise $n \geq 30$, so the Central Limit Theorem makes $\bar{x}$ approximately normal	$z_{obs} \sim N(0,1)$
t-test for μ (σ unknown)	approximately normal, $X \sim N(\mu, \sigma)$	any n if the population is normal; otherwise $n \geq 30$	$t_{obs} \sim t(n-1)$

If an assumption fails, the curve we read the P-value from is the wrong curve, and the P-value can be wrong. With a large sample the Central Limit Theorem usually rescues the tests for a mean; with a small sample from a population we cannot check, nothing does.

When those assumptions are doubtful, and especially when the sample is too small to check them, we need another option. A **non-parametric** test makes few or no assumptions about the shape of the population's distribution. It still assumes the data come from a random sample (or a randomized experiment) and that the observations are independent.

- **Advantages of non-parametric tests**
    1. They may be the only option when the sample is very small and the population's distribution is unknown, which is almost always.
    2. They make few assumptions about the population.
    3. They suit crude measurements, such as subjective ratings or rankings (for example, Likert responses).
    4. Their computations and interpretations are often simpler.
- **Disadvantage:** they are generally **less powerful** than the parametric test when that test's assumptions hold. They throw away some of the information in the data, so they need more evidence to reject H_0.

The sign test

The **sign test** is a non-parametric test for a population **median**. The t-test asks about the mean, and needs a normal population (or a large sample) to do it; the sign test asks about the median and needs no assumption about the population's shape.

We use it here for **matched pairs**: two measurements on each subject (before and after a treatment) or on each matched pair (a husband and a wife). The parameter is the **median difference**, M_D, the median of all the differences in the population of pairs. For each pair in the sample we find the difference and keep only its **sign**, $+$ or $-$, ignoring its size. A difference of exactly zero has no sign, so we drop it; n is the number of **non-zero** differences.

The parameter and the hypotheses

"No systematic difference between the two measurements" means the median difference is zero:

$$H_0 : M_D = 0$$

The sign test works because of what a median is: **half of any population lies above its**

median and half below. So if $M_D = 0$, a difference is just as likely to be positive as negative. Writing

$$p = P(\text{a difference is } +),$$

the null hypothesis about the median is the same as a null hypothesis about this proportion:

$$H_0 : M_D = 0 \iff H_0 : p = 0.5$$

If the median difference is above 0, more than half of the differences are positive ($p > 0.5$), and if it is below 0, fewer are. The alternative depends on the question:

Alternative	Name	Meaning
$H_A : M_D > 0$, so $p > 0.5$	upper-tailed	the first measurement tends to be larger: more $+$ signs than chance
$H_A : M_D < 0$, so $p < 0.5$	lower-tailed	the first measurement tends to be smaller: more $-$ signs than chance
$H_A : M_D \neq 0$, so $p \neq 0.5$	two-sided	a systematic difference in either direction

The same idea tests the median of a **single** population: to test $H_0 : M = M_0$, give each observation the sign of $X - M_0$ and count the $+$ signs. Matched pairs are the special case $M_0 = 0$ applied to the differences.

The test statistic and its distribution

The test statistic is

$$S = \text{the number of } + \text{ signs among the } n \text{ non-zero differences}$$

Recall from Week 7 that the number of successes in n independent trials, each with the same chance of success, has a binomial distribution. Here each non-zero difference is a trial and a $+$ is a success, so if H_0 is true

$$S \sim \text{Binomial}(n, 0.5), \qquad P(S = k) = \frac{n!}{k!\,(n-k)!}\,(0.5)^k (0.5)^{n-k} = \frac{n!}{k!\,(n-k)!}\,(0.5)^n$$

Like any binomial count, its mean and standard error are np and $\sqrt{np(1-p)}$, with $p = 0.5$:

$$E(S) = \frac{n}{2}, \qquad SE(S) = \sqrt{n \times 0.5 \times 0.5} = \frac{\sqrt{n}}{2}$$

So if H_0 is true we expect about half the signs to be $+$, give or take $\sqrt{n}/2$.

Large samples. When $n/2 \geq 15$ (the same $np_0 \geq 15$ check as the test for a proportion, with $p_0 = 0.5$), S is approximately normal and we can standardise it:

$$z_{obs} = \frac{S - n/2}{\sqrt{n}/2} \sim N(0,1) \text{ if } H_0 \text{ is true}$$

This is exactly the z-test for a proportion with $p_0 = 0.5$: dividing the top and bottom by n gives $z_{obs} = (\hat{p} - 0.5)/\sqrt{0.5 \times 0.5/n}$, with $\hat{p} = S/n$.

Small samples. When $n/2 < 15$, use the binomial probabilities exactly.

The P-value

Alternative	Exact (small n)	Large n (z-table)
$H_A : p > 0.5$	$P(S \geq s \mid H_0 \text{ true})$: add $P(S = k)$ for $k = s, \dots, n$	$P(Z \geq z_{obs})$
$H_A : p < 0.5$	$P(S \leq s \mid H_0 \text{ true})$: add $P(S = k)$ for $k = 0, \dots, s$	$P(Z \leq z_{obs})$
$H_A : p \neq 0.5$	2× the smaller of the two tails	$2 \times P(Z \geq \lvert z_{obs} \rvert)$

The five steps

1. **Assumptions.** Matched pairs; the pairs are independent of each other. Nothing is assumed about the shape of the population.
2. **Hypotheses.** $H_0 : M_D = 0$ against $H_A : M_D > 0$, $M_D < 0$ or $M_D \neq 0$; in terms of $p = P(+)$, $H_0 : p = 0.5$ against $p > 0.5$, $p < 0.5$ or $p \neq 0.5$.
3. **Test statistic.** Find each difference and its sign, drop the zeros, and count S, the number of $+$ signs among the n that remain. Compare it with $E(S) = n/2$; if $n/2 \geq 15$, also find z_{obs}.
4. **P-value.** From $S \sim \text{Binomial}(n, 0.5)$ (or the z-table, for large n), in the tail(s) H_A points to.
5. **Decision.** Reject H_0 if the P-value is at most α.

Want to learn more? The sign test throws away the size of each difference. The **Wilcoxon signed-rank test** is another test for the median difference of matched pairs that keeps some of that information by ranking the absolute differences, and the **Wilcoxon rank-sum test** compares the medians of two independent samples. Both are worked through, with examples, in Non-parametric tests for two samples (misc/Non-Parametric-Tests-For-Two-Samples.html), under **Resources** → **Extras**. This is optional reading.

Example: who made the decision?

A study asked married couples who had more say in a recent major financial decision. Each husband and wife answered separately, on a scale from 1 (the wife had more say) to 7 (the husband had more say). Do husbands and wives see these decisions the same way, or is there a systematic difference? Here are 17 couples:

Couple	Husband	Wife	Difference	Sign
1	5	3	2	+
2	4	3	1	+
3	6	4	2	+
4	6	5	1	+
5	3	3	0	
6	2	3	-1	−
7	5	2	3	+
8	3	3	0	
9	1	2	-1	−
10	4	3	1	+
11	5	2	3	+
12	4	2	2	+
13	4	5	-1	−
14	7	2	5	+

Couple	Husband	Wife	Difference	Sign
15	5	5	0	
16	5	3	2	+
17	5	1	4	+

A t-test on the differences would assume they come from a normal population, and that is hard to believe here: there are only 17 couples, and the ratings are crude, ordered scores from 1 to 7. The sign test asks only whether each difference is $+$ or $-$.

1. **Assumptions.** Matched pairs (a husband and wife from each couple), and the couples are independent.
2. **Hypotheses.** M_D = the median of husband $-$ wife over all couples. $H_0 : M_D = 0$, no systematic difference; $H_A : M_D \neq 0$, **two-sided**, since the question has no direction. Equivalently, with p = the probability that the husband rates his say higher (a $+$ difference), $H_0 : p = 0.5$ against $H_A : p \neq 0.5$. Use $\alpha = 0.05$.
3. **Test statistic.** 11 differences are $+$, 3 are $-$, and 3 are zero. Dropping the zeros, $n = 14$ and $S = 11$. If H_0 is true,

$$E(S) = \frac{14}{2} = 7, \qquad SE(S) = \frac{\sqrt{14}}{2} \approx 1.87$$

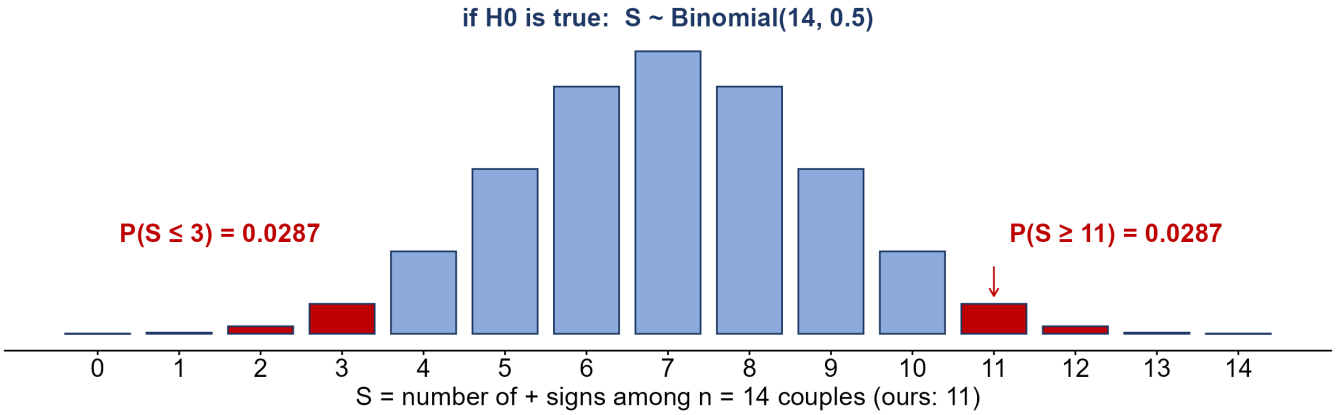
   so 11 is about 2.1 standard errors above what we would expect. But $n/2 = 7 < 15$, so the normal approximation is not good enough here; we use the binomial probabilities exactly.
4. **P-value.** If H_0 is true, $S \sim \text{Binomial}(14, 0.5)$. The upper tail is

$$P(S \geq 11) = \frac{\binom{14}{11} + \binom{14}{12} + \binom{14}{13} + \binom{14}{14}}{2^{14}} = \frac{364 + 91 + 14 + 1}{16{,}384} = \frac{470}{16{,}384} \approx 0.0287$$

   The test is two-sided, so we double it:

$$\text{P-value} = 2 \times P(S \geq 11 \mid H_0 \text{ true}) = 2 \times 0.0287 = 0.0574$$

5. **Decision.** $0.0574 > 0.05$: **do not reject** H_0. There is not enough evidence that the median difference is different from 0: husbands and wives do not differ significantly in how much say they think each had, at the 5% level. It is close, though: a one-sided test, or a larger sample, might have rejected.



Practice: a stretching program

A hypothetical example. A physical therapist thinks a six-week stretching program **lowers** knee pain. Ten patients with chronic knee pain rate their pain from 0 (none) to 10 (worst) before and after the program. The difference is before — after, so a + means the patient's pain went down:

Patient	Before - after	Sign
1	3	+
2	1	+
3	0	
4	2	+
5	4	+
6	-1	−
7	2	+
8	1	+
9	3	+
10	2	+

Carry out a sign test at $\alpha = 0.05$.

Answer.

1. **Assumptions.** Matched pairs (each patient before and after); the patients are independent. Pain ratings are crude scores and n is small, so the sign test suits them.
2. **Hypotheses.** M_D = the median of before − after over all such patients. $H_0 : M_D = 0$ against $H_A : M_D > 0$ (equivalently $p > 0.5$): **upper-tailed**, since the therapist expects pain to go down.
3. **Test statistic.** 8 +, 1 − and 1 zero, so $n = 9$ and $S = 8$. If H_0 is true, $E(S) = 4.5$ and $SE(S) = \sqrt{9}/2 = 1.5$; $n/2 = 4.5 < 15$, so use the exact binomial.
4. **P-value.** With $S \sim \text{Binomial}(9, 0.5)$,

$$P(S \geq 8 \mid H_0 \text{ true}) = \frac{\binom{9}{8} + \binom{9}{9}}{2^9} = \frac{9+1}{512} \approx 0.0195$$

5. **Decision.** $0.0195 < 0.05$: **reject** H_0. The data are evidence that the median change is positive: the program lowers knee pain.

Practice: a new coffee blend

A hypothetical example. A coffee shop thinks its regular customers prefer a new blend to the old one. It asks **60** regulars to taste both and rate each from 1 to 10. The difference is new − old, so a + means the customer rated the new blend higher. **38** differences are +, **16** are −, and **6** are ties. Carry out a sign test at $\alpha = 0.05$.

Answer.

1. **Assumptions.** Matched pairs (each customer rates both blends); the customers are independent. The ratings are crude scores, so the sign test suits them.
2. **Hypotheses.** M_D = the median of new − old over all regulars. $H_0 : M_D = 0$ against $H_A : M_D > 0$ (equivalently $p > 0.5$): **upper-tailed**, since the shop expects the new blend to rate higher.
3. **Test statistic.** Drop the 6 ties: $n = 54$ and $S = 38$. If H_0 is true,

$$E(S) = \frac{54}{2} = 27, \qquad SE(S) = \frac{\sqrt{54}}{2} \approx 3.674$$

   Now $n/2 = 27 \geq 15$, so we can use the normal approximation:

$$z_{obs} = \frac{38 - 27}{3.674} \approx 2.99$$

4. **P-value.** $P(Z \geq 2.99 \mid H_0 \text{ true}) = 1 - 0.9986 = 0.0014$. (The exact binomial tail, $P(S \geq 38)$, is 0.0019; the approximation is close.)

5. **Decision.** $0.0014 < 0.05$: **reject** H_0. The data are evidence that the median difference is positive: regulars tend to rate the new blend higher.

Sources

- All examples are from OpenStax *Introductory Statistics 2e*, Chapter 9: the bench press (Example 9.15), glass conductivity (Example 9.20), the baker's bread (Example 9.9 and Try It 9.9), the composite hypotheses (Table 9.1 and Examples 9.1, 9.3 and 9.8), the errors (Table 9.2, Examples 9.5, 9.7 and 9.8, Try It 9.6), cell phone owners (Try It 9.18), and the goggles (Example 9.14) and Exercises 17 and 19.
- The goggles' true mean of 16.0 seconds in the power example is an assumption for illustration.
- The couples' decision ratings come from the 2024 STAT 251 notes; the stretching program and the coffee blend are hypothetical. The textbook does not cover the sign test.
- Textbook: OpenStax *Introductory Statistics 2e*, Sections 9.1–9.5.